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SYSTEM AND METHOD FOR SHIFTING TWO SPEED TRANSMISSION

Abstract

Methods and systems for shifting a transmission that includes a dog clutch are described. In one example, the transmission may be coupled to an electric machine and a rotational speed of the electric machine may be adjusted to reduce a speed differential between rotating transmission components. A position of a shifter actuator may be adjusted according to an estimate of when the speed differential is equal to a predetermined value.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to shifting a two speed transmission of an electric vehicle.

BACKGROUND AND SUMMARY

[0002] Off-highway battery electric vehicles may operate with large speed and load ranges. For example, a rock truck may carry a large load that requires a large amounts of torque for the increasing the rock truck's speed from rest. However, at times, the rock truck may be empty and it may be desirable to move it from one location to another location in a short period of time. Therefore, it may be desirable for the rock truck to reach higher speeds. A single speed transmission in the off-highway battery electric vehicle may not make it possible for the off-highway battery electric vehicle to operate within a desired speed and load operating range. Therefore, a two speed transmission may be installed in the off-highway battery electric vehicle. The two speed transmission may allow the vehicle to operate at higher vehicle speed while generating sufficient torque to propel a vehicle that is highly loaded at lower vehicle speeds. Nevertheless, shifting the two speed transmission may not be as simple as may be desired because electric machines may rotate at relatively high speeds that may result in a large speed gap between rotational speed of a first transmission shaft (e.g., an input shaft) and rotational speed of a second transmission shaft (e.g., an output shaft). The large speed gap between transmission shaft rotational speeds may make it more difficult for synchronizers to equalize the speed between the two shafts so that a shift may be brought about. Consequently, the amount of time for a shift to occur may be extended longer than may be desired and vehicle drivability may degrade.

[0003] The inventors herein have recognized the above-mentioned issues and have developed an electric propulsion system, comprising: an electric machine; a step gear ratio transmission including at least two gear ratios, the transmission including a shifter actuator; and one or more controllers including executable instructions that cause the one or more controllers to estimate an amount of time for a differential speed between a rotational speed of an on-coming gear and a rotational speed of a transmission shaft to be less than a threshold rotational speed.

[0004] By estimating an amount of time it takes for a differential speed between two transmission components to be equal to or less than a predetermined amount, it may be possible to provide the technical result of coordinating shifter movement with component speed synchronization so that an amount of time it takes to shift gears of a transmission may be reduced. In addition, a dog clutch may be engaged smoothly since the approach is based on rotational speeds of components in the transmission being synchronized.

[0005] The present description may provide several advantages. In particular, the approach may make reduce an amount of time it takes to shift transmission gears. Further, the approach may allow a shifter to engage a dog clutch so that a possibility of shift shock may be reduced. Additionally, the approach may reduce dog clutch wear.

[0006] It is to be understood that the summary above is provided to introduce in simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key features of the claimed subject matter, the scope of which is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not restricted to implementations that solve any disadvantages noted above or in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is an illustration of an example vehicle that includes an electric vehicle propulsion system.

[0008] FIG. 2 shows a stick diagram of one example two gear ratio transmission configuration.

[0009] FIG. 3 shows an example dog clutch.

[0010] FIG. 4 shows operating states of a dog clutch during a gear shift sequence.

[0011] FIG. 5 shows signals of interest during a shifting sequence according to the present

description.

[0012] FIG. **6** shows a differential speed profile between transmission shaft speeds during transmission gear shifting.

[0013] FIG. **7** shows a graphic of how a differential speed between two transmission shafts may be predicted.

[0014] FIGS. **8-14** show plan views of example teeth of a dog clutch in a range of positions.

[0015] FIG. **15** shows a flow chart of an example method for shifting gears of a transmission.

[0016] FIG. **16** shows a flow chart of an example method for operating a gear shift actuator of a transmission.

DETAILED DESCRIPTION

[0017] A method and system for shifting a transmission is described. The transmission may be a two speed transmission (e.g., a transmission with two forward gear ratios) or a transmission having more than two gear ratios. The transmission shifting may include predicting a speed differential and an amount of time to move a shifter so that shift delay may be reduced. By reducing the shift delay, the transmission may be operated in a state where torque is not transferred through the transmission for a shorter period of time, thereby increasing torque delivery to the vehicle's wheels. The transmission may be included in an electric vehicle as shown in FIG. **1**. The transmission may be of the type shown in FIG. **2** and having a dog clutch as shown in FIG. **3** that operates as shown in FIGS. **4-14**. A method for shifting gears of a transmission is shown in FIG. **15** and a method for controlling a gear shift actuator is shown in FIG. **16**.

[0018] FIG. **1** illustrates an example vehicle propulsion system **199** for vehicle **10**. In FIG. **1** mechanical connections between the various components are illustrated as solid lines, whereas electrical connections between various components are illustrated as dashed lines. Vehicle front end is indicated at **110** and vehicle rear end is indicated at **111**. Vehicle **10** travels in a forward direction when vehicle front end **110** leads movement of vehicle **10**. Vehicle **10** travels in a reverse direction when vehicle rear end **111** leads movement of vehicle **10**. In this example, vehicle **10** is a rear wheel drive vehicle, but in other examples, vehicle **10** may be a four-wheel drive or front wheel drive vehicle.

[0019] Vehicle propulsion system **199** includes a propulsion source **105** (e.g., an electric machine, such as a motor), but in other examples two or more propulsion sources may be provided. In one example, propulsion source **105** may be a synchronous or induction electric machine that may operate as a motor or generator. In other examples, propulsion source **105** may be a direct current (DC) machine. Vehicle propulsion system **199** also includes a transmission **135**. The propulsion source **105** is fastened to the transmission **135** and propulsion source **105** delivers power from its rotor **105a** to transmission **135**. Transmission **135** may be mechanically coupled to differential gears. Differential gears **106** may be coupled to two axle shafts, including a first or right axle shaft **190a** and a second or left axle shaft **190b**. Vehicle **10** further includes front wheels **102** and rear wheels **103**.

[0020] The transmission **135** may be referred to as a step ratio transmission and it may be configured as shown in greater detail in FIG. **2**, or alternatively, a different configuration. Transmission **135** may include one or more clutch actuators (not shown) to shift one or more clutches. Transmission **135** may include a first speed sensor **119** for sensing a rotational speed of propulsion source **105** and a second speed sensor **185** for sensing a transmission output shaft speed. Electric power inverter **115** is electrically coupled to propulsion source **105** to convert DC power to alternating current (AC) and vice-versa. Powertrain controller **116** is electrically coupled to sensors **117** and actuators of vehicle propulsion system **199**. For example, sensors **117** may include, but are not limited to inverter switch temperature sensors, electric machine winding temperature sensors, bus bar temperature sensors, etc.

[0021] Transmission **135** may transfer mechanical power to or receive mechanical power from differential gears **106**. Differential gears **106** may transfer mechanical power to or receive

mechanical power from rear wheels **103** via right axle shaft **190a** and left axle shaft **190b**. Propulsion source **105** may consume alternating current (AC) electrical power provided via electric power inverter **115**. Alternatively, propulsion source **105** may provide AC electrical power to electric power inverter **115**. Electric power inverter **115** may be provided with high voltage direct current (DC) power from battery **160** (e.g., a traction battery, which also may be referred to as an electric energy storage device or battery pack). Electric power inverter **115** may convert the DC electrical power from battery **160** into AC electrical power for propulsion source **105**. Alternatively, electric power inverter **115** may be provided with AC power from propulsion source **105**. Electric power inverter **115** may convert the AC electrical power from propulsion source **105** into DC power to store in battery **160**.

[0022] Propulsion source **105** may transfer mechanical power to or receive mechanical power from transmission **135**. As such, transmission **135** may be a multi-speed gear set that may shift between gear ratios when commanded via powertrain controller **116**. Powertrain controller **116** includes a processor **116a** and memory **116b**. Memory **116b** (e.g., storage media) may include read exclusive memory, random access memory, and keep alive memory. The memory may be programmed with computer readable data representing instructions that are executable by a processor for performing the methods and control techniques described herein as well as other variants that are anticipated but not specifically listed. As such, control techniques, methods, and the like expanded upon herein may be stored as instructions in non-transitory memory.

[0023] Battery **160** may periodically receive electrical energy from a power source such as a stationary power grid **5** residing external to the vehicle (e.g., not part of the vehicle). As a non-restricted example, vehicle propulsion system **199** may be configured as a plug-in electric vehicle (EV), whereby electrical energy may be supplied to battery **160** via the stationary power grid **5** and charging station **12**. Electric charge may be delivered to battery **160** via plug receptacle **100**.

[0024] Battery **160** may include a BMS controller **139** (e.g., a battery management system controller) and an electrical power distribution box **162**. BMS controller **139** may provide charge balancing between energy storage elements (e.g., battery cells) and communication with other vehicle controllers (e.g., vehicle control unit **152**). BMS controller **139** includes a core processor **139a** and memory **139b** (e.g., random-access memory, read-exclusive memory, and keep-alive memory).

[0025] Vehicle **10** may include a vehicle control unit (VCU) **152** that may communicate with electric power inverter **115**, powertrain controller **116**, friction or foundation caliper controller **170**, global positioning system (GPS) **188**, BMS controller **139**, and dashboard **186** and components included therein via controller area network (CAN) **120**. VCU **152** includes memory **114**, which may include read-exclusive memory (ROM or non-transitory memory) and random access memory (RAM). VCU also includes a digital processor or central processing unit (CPU) **153**, and inputs and outputs (I/O) **118** (e.g., digital inputs including counters, timers, and discrete inputs, digital outputs, analog inputs, and analog outputs). VCU may receive signals from sensors **154** and provide control signal outputs to actuators **156**. Sensors **154** may include but are not restricted to lateral accelerometers, longitudinal accelerometers, yaw rate sensors, inclinometers, temperature sensors, battery voltage and current sensors, and other sensors described herein. Additionally, sensors **154** may include steering angle sensor **197**, driver demand pedal position sensor **141**, vehicle range finding sensors including radio detection and ranging (RADAR), light detection and ranging (LIDAR), sound navigation and ranging (SONAR), and caliper application pedal position sensor **151**. Actuators may include but are not constrained to inverters, transmission controllers, display devices, human/machine interfaces, friction caliper systems, and battery controller described herein.

[0026] Driver demand pedal position sensor **141** is shown coupled to driver demand pedal **140** for determining a degree of application of driver demand pedal **140** by human **142**. Caliper application pedal position sensor **151** is shown coupled to caliper application pedal **150** for determining a

degree of application of caliper application pedal **150** by human **142**. Steering angle sensor **197** is configured to determine a steering angle according to a position of steering wheel **198**.

[0027] Vehicle propulsion system **199** is shown with a global position determining system **188** that receives timing and position data from one or more GPS satellites **189**. Global positioning system may also include geographical maps in ROM for determining the position of vehicle **10** and features of roads that vehicle **10** may travel on.

[0028] Vehicle propulsion system **199** may also include a dashboard **186** that an operator of the vehicle may interact with. Dashboard **186** may include a display system **187** configured to display information to the vehicle operator. Display system **187** may comprise, as a non-restricting example, a touchscreen, or human machine interface (HMI), display which enables the vehicle operator to view graphical information as well as input commands. In some examples, display system **187** may be connected wirelessly to the internet (not shown) via VCU **152**. As such, in some examples, the vehicle operator may communicate via display system **187** with an internet site or software application (app) and VCU **152**.

[0029] Dashboard **186** may further include an operator interface **182** via which the vehicle operator may adjust the operating status of the vehicle. Specifically, the operator interface **182** may be configured to activate and/or deactivate operation of the vehicle driveline (e.g., propulsion source **105**) based on an operator input. Further, an operator may request an axle mode (e.g., park, reverse, neutral, drive) via the operator interface. Various examples of the operator interface **182** may include interfaces that utilize a physical apparatus, such as a key, that may be inserted into the operator interface **182** to activate the vehicle propulsion system **199** including propulsion source **105** and to turn on the vehicle **10**. The apparatus may be removed to shut down the transmission **135** and propulsion source **105** to turn off vehicle **10**. Propulsion source **105** may be activated via supplying electric power to propulsion source **105** and/or electric power inverter **115**. Propulsion source **105** may be deactivated by ceasing to supply electric power to propulsion source **105** and/or electric power inverter **115**. Still other examples may additionally or optionally use a start/stop button that is manually pressed by the operator to start or shut down the propulsion source **105** to turn the vehicle on or off. In other examples, a remote electrified axle or electric machine start may be initiated remote computing device (not shown), for example a cellular telephone, or smartphone-based system where a user's cellular telephone sends data to a server and the server communicates with the vehicle control unit **152** to activate the inverter **115** and propulsion source **105**. Spatial orientation of vehicle **10** is indicated via axes **175**.

[0030] Vehicle **10** is also shown with a foundation or friction caliper controller **170**. Friction caliper controller **170** may selectively apply and release friction calipers (e.g., **172a** and **172b**) via allowing hydraulic fluid to flow to the friction calipers. The friction calipers may be applied and released so as to reduce locking of the friction calipers to front wheels **102** and rear wheels **103**. Wheel position or speed sensors **161** may provide wheel speed data to friction caliper controller **170**. Vehicle propulsion system **199** may provide torque to rear wheels **103** to propel vehicle **10**.

[0031] A human or autonomous driver **142** may request a driver demand wheel torque, or alternatively a driver demand wheel power, via applying driver demand pedal **140** or via supplying a driver demand wheel torque/power request to vehicle control unit **152**. Vehicle control unit **152** may then demand a torque or power from propulsion source **105** via commanding powertrain controller **116**. Powertrain controller **116** may command electric power inverter **115** to deliver the driver demand wheel torque/power via electrified axle **190** and propulsion source **105**. Electric power inverter **115** may convert DC electrical power from battery **160** into AC power and supply the AC power to propulsion source **105**. Propulsion source **105** rotates and transfers torque/power to transmission **135**. Transmission **135** may supply torque from propulsion source **105** to differential gears **106**, and differential gears **106** transfer torque from propulsion source **105** to rear wheels **103** via axle shafts **190a** and **190b**.

[0032] During conditions when the driver demand pedal is fully released, vehicle control unit **152**

may request a small negative or regenerative power to gradually slow vehicle **10** when a speed of vehicle **10** is greater than a threshold speed. The amount of regenerative power requested may be a function of driver demand pedal position, battery state of charge (SOC), vehicle speed, and other conditions. If the driver demand pedal **140** is fully released and vehicle speed is less than a threshold speed, vehicle control unit **152** may request a small amount of positive torque/power (e.g., propulsion torque) from propulsion source **105**, which may be referred to as creep torque or power. The creep torque or power may allow vehicle **10** to remain stationary when vehicle **10** is on a small positive grade.

[0033] The human or autonomous driver may also request a negative or regenerative driver demand slowing torque, or alternatively a driver demand slowing power, via applying caliper pedal **150** or via supplying a driver demand slowing power request to vehicle control unit **152**. Vehicle control unit **152** may request that a first portion of the driver demanded slowing power be generated via propulsion source **105** via commanding powertrain controller **116**. Additionally, vehicle control unit **152** may request that a portion of the driver demanded slowing power be provided via friction calipers **172a** and **172b** via commanding friction caliper controller **170** to provide a second portion of the driver requested slowing power.

[0034] After vehicle control unit **152** determines the slowing power request, vehicle control unit **152** may command powertrain controller **116** to deliver the portion of the driver demand slowing power allocated to propulsion source **105**. Propulsion source **105** may convert the vehicle's kinetic energy into AC power.

[0035] Powertrain controller **116** includes predetermined transmission gear shift schedules whereby fixed ratio gears of transmission **135** may be selectively engaged and disengaged. Shift schedules stored in powertrain controller **116** may select gear shift points or events as a function of driver demand wheel torque and vehicle speed.

[0036] Turning now to FIG. 2, a stick diagram of transmission **135** is shown. In this example, transmission **135** is a step transmission with two gear ratios (e.g., a first lower gear ratio **206** and a second higher gear ratio **204**). In other examples, transmission **135** may include additional gear ratios.

[0037] Transmission **135** includes input shaft **202** for receiving mechanical torque from propulsion source **105**. Propulsion source **105** is supplied with electric power via inverter **115** and a traction battery (not shown). Rotational speed of propulsion source **105** and input shaft **202** may be determined via first speed sensor **119**. Gear **250** and gear **254** are fixed to input shaft **202** and these gears rotate at a same speed as input shaft **202**. Gear **250** is meshed with second idler gear **252** and gear **254** is meshed with first idler gear **256**. Gear **250** and gear **252** along with gears **258** and **260** may form a second higher gear ratio (e.g., second gear). Gear **254** and gear **256** along with gears **258** and **260** may form a first lower gear ratio (e.g., first gear). Gears **252** and **256** may rotate freely about intermediate or layshaft **212** when dog clutch **210** is not engaged. Dog clutch **210** may selectively engage either gear **252** to engage the second higher gear or selectively engaged gear **256** to engage the first lower gear. Torque flow through transmission is interrupted when dog clutch **210** is not engaged to either of gears **252** and **256**. Dog clutch **210** rotates with intermediate shaft so that when dog clutch **210** engages gear **252**, gear **252** rotates at a same speed as intermediate shaft **212**. Likewise, dog clutch **210** may engage gear **256** so that gear **256** rotates at a same speed as intermediate shaft **212**. Intermediate shaft **212** rotates at a multiple of a rotational speed of output shaft **230**. Intermediate shaft **212** includes gear **258** that meshes with gear **260** that is coupled to output shaft **230**. Output shaft may rotate differential **106**. Dog clutch **210** may be moved axially with respect to intermediate shaft **212** as indicated by arrow **255** via shift fork **214**. Shift fork **214** may be move axially via gear shift actuator **220**. Controller **116** may adjust a position of gear shift actuator **220** via adjusting hydraulic valve **222** and hydraulic valve **224**. A position of shift fork **214** and a position of dog clutch **210** may be determined via shift position sensor **216**. The positions of the speed sensors for the transmission may be different than those shown in FIG. 2 without

affecting operation and control of transmission **135**.

[0038] FIG. **3** shows an example dog clutch **210** of vehicle propulsion system **199**. In FIG. **3**, a gear **252** that may be selectively engaged by dog clutch **210** is shown. The other gear (e.g., gear **256**) that may be engaged by dog clutch **210** is not shown for clarity. It may be appreciated that other dog clutch configurations are also possible.

[0039] Dog clutch **210** is a locking device that may selectively lock gear **252** to intermediate shaft **212** to engage a transmission gear (e.g., second gear). Gear **252** may freely rotate independent of the rotation of intermediate shaft **212** via needle bearing (not shown) that are between gear **252** and intermediate shaft **212**. Hub **303** includes an inner spline **307** that is permanently mated with the spline **305** of gear **252** so that hub **303** is rotatably connected to shaft **212**. Hub **303** is also axially fixed to shaft **212** via a couple of circlips (not shown). Sleeve **308** is rotatably connected to hub **303** by spline teeth pairs **304** and **309**. Sleeve **308** may slide axially across hub **303**. A fork (not shown) may drive axially sleeve **308** via groove **310**. When sleeve **308** is urged towards gear **252**, spline teeth **309** couples with both splines **306** and **304**, thereby rotatably connecting gear **252** and shaft **212**. Spline **306** may be referred to as engagement ring or locking ring. Sleeve **308** may travel towards gear **252** up to a position where end-stop **311** reaches tip of spline **306**. A control strategy may adjust a speed of gear **252** within a close range of a speed of shaft **212** so that dog clutch may be engaged with minimal interference of teeth. In a strategy, synchronization is the operation of matching rotational speed of hub **303**, which may be imposed by vehicle speed, with rotational speed of spline **306**, which may be controlled via the traction motor. This operation may be referred to as electrical synchronization. Often, sleeve **310** moving apart from gear **301** will approach a gear of another ratio passing through a neutral position. This second gear is not represented for the sake of clarity.

[0040] Referring now to FIG. **4**, operating states of dog clutch teeth for a sequence where a first gear is disengaged and a second gear is engaged via a dog clutch is shown. The sequence begins on the left side of the figure and it ends on the right side of the figure. Table 400 indicates dog clutch engagement reliability levels and engagement depths from disengagement of gear **252** to engagement of gear **256**.

[0041] Spline teeth **306** of gear **252** are indicated as polygons and spline teeth **406** of gear **256** are indicated as polygons. Spline teeth **309** of dog clutch **210** are shown positioned between spline teeth **306** and spline teeth **406** for each operating state. Each of the operating states is indicated by a group of spline teeth **306**, spline teeth **309** of dog clutch **210**, and a group of spline teeth **406**.

[0042] At the left most side of FIG. **4**, the spline teeth **309** of dog clutch **210** are fully engaged with the spline teeth **306** such that gear **252** (not shown) is locked to the intermediate shaft (not shown). Moving to the right, as gear **252** (not shown) is disengaged, spline teeth **309** of dog clutch **210** clear spline teeth **306** of gear **252** such that the transmission enters a neutral state. As the spline teeth **309** of dog clutch **210** (not shown) move right, the dog clutch moves from the neutral state to a repulsive state before becoming fully engaged at the right side of FIG. **4**. There is substantial overlap between spline teeth **309** and spline teeth **406** when gear **256** is fully engaged.

[0043] Moving on to FIG. **5**, a sequence of disengaging a first gear and engaging a second gear according to the method of FIG. **16** is shown. The sequence may be generated via the system of FIGS. **1-3** in cooperation with the method of FIG. **166**.

[0044] The first plot from the top is a plot of shift actuator position versus time. The vertical axis indicates shifter position when engaging a first gear above axis and shifter position when engaging a second gear below the axis. The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Solid line **502** represents actual actuator position and dashed line **504** represents target or requested actuator position.

[0045] The second plot from the top is a plot of a rotational speed difference between rotational speed of the disengaged gear and intermediate shaft rotational speed versus time. The vertical axis represents the rotational speed difference between the rotational speed of a disengaged gear (e.g.,

on-coming gear) and a rotational speed of the intermediate shaft. The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Solid line **506** represents the rotational speed difference between rotational speed of the disengaged gear and intermediate shaft rotational speed versus time.

[0046] The third plot from the top is a plot of an estimated gear engagement time versus time. The vertical axis represents the estimated gear engagement time for the gear shift. The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Solid line **508** represents the estimated gear engagement time for the gear shift. Dashed line **550** represents an amount of time it takes to move the shifter and dog clutch sleeve far enough to achieve a stable torque transfer capacity (e.g., the dog clutch having capacity to transfer torque demanded by the vehicle driver).

[0047] The fourth plot from the top is a plot of a synchronous phase state for a gear shift versus time. The vertical axis represents the synchronous phase state for the gear shift. The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Solid line **510** represents the synchronous phase state for the gear shift.

[0048] The fifth plot from the top is a plot of an engagement phase state for a gear shift versus time. The vertical axis represents the engagement phase state for the gear shift. The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Solid line **512** represents the engagement phase state for the gear shift.

[0049] At time t_0 , the first gear is engaged and the shift actuator is in a fully engaged position. The speed difference is zero and the estimated engagement time is zero. The gear shift has not started so the synchronous phase state is not asserted and the engagement phase is not asserted.

[0050] At time t_1 , a gear shift is initiated and a disengagement phase of the presently engaged gear (e.g., the off-going gear) begins to be release by adjusting the position of the shifter shortly after time t_1 . The rotational speed difference between the speed of the intermediate shaft and the rotational speed of the on-coming gear jumps up to a higher value. The synchronization phase is not active and the engagement phase is not active.

[0051] At time t_2 , the shift actuator has passed through its mid-point or center of travel. The target shifter position advances toward engaging the second gear. The rotational speed difference between the on-coming gear and the intermediate shaft is unchanged. The synchronous phase begins and the electric machine speed is adjusted to drive the rotational speed of the on-coming gear toward the rotational speed of the intermediate shaft. The estimated engagement time begins to be calculated shortly after time t_2 and the estimated engagement time declines as time progresses and the rotational speed of the on-coming gear moves toward the rotational speed of the intermediate shaft. The engagement phase is state is not asserted.

[0052] At time t_3 , the estimated amount of time it takes for the rotational speed difference between the on-coming gear and the intermediate shaft to reach zero or a predefined value is less than or equal to the time it takes to move the shifter, shift fork, and sleeve to the engaged position. Therefore, the synchronization phase ends and the engagement phase begins. The rotational speed difference between the rotational speed of the on-coming gear and the intermediate shaft rotational speed continues to decline. The target shift actuator position changes to a full engagement position and the actual shifter position follows the target after a short lag period. The engagement phase ends shortly after time t_3 when the actual shifter position matches the target shifter position.

[0053] Referring now to FIG. 6, a plot of an example desired speed differential or speed difference profile during a synchronous phase of a transmission gear shift is shown. The vertical axis represents a rotational speed difference between a rotational speed of an intermediate shaft and a rotational speed of an on-coming gear (e.g., a gear that is in the process of being engaged). The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Trace **602** represents the rotational speed difference between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear.

[0054] At a time of zero, the rotational speed difference between the intermediate shaft rotational speed and the on-coming gear rotational speed is at a high value. The electric machine speed is adjusted by adjusting torque of the propulsion source to near or at maximum torque of the propulsion source so that the rotational speed difference is reduce quickly. As the rotational speed difference approaches zero, the electric machine torque is adjusted further to reduce the rate of rotational differential speed change. In one example, upon approaching a target differential speed, the control torque adjustments become less substantial and they may resemble a first order closed-loop transfer function of the form:

$$[00001] \frac{d(s)}{d_{\text{target}}(s)} \cong \frac{1}{1 + s}$$

where $\omega.\text{sub}.d$ is the rotational speed difference between the rotational speed of the intermediate shaft rotational speed and the on-coming gear rotational speed, s is a Laplace complex variable, and $\omega.\text{sub}.d,\text{target}$ is a target rotational speed difference.

[0055] The motion equation of the adrift vehicle forced by rolling resistance and track slope is a first order one in velocity (namely $\{\dot{v}\}m=F$ where m is the vehicle mass and F is the sum of all the applied forces and $\{\dot{v}\}$ is rate of vehicle speed change) as well as motion equation for the propulsion source, disconnected from the load, is a first order one in angular velocity, namely $\omega J=T$, where J is the angular mass of the rotor, T is propulsion source torque, and ω is the propulsion source rate of angular velocity change and of the transmission elements that are permanently connected to the propulsion source (e.g., components upstream the dog clutch). The differential speed $\omega.\text{sub}.d$ is a linear combination of vehicle velocity and motor angular velocity, both first order systems as seen, its dynamics resemble a first order system. This effect is visible in FIG. 6 where the transition from a closed-loop response working in torque saturation to a less aggressive speed control is marked by an asterisk. The system time constant t is lower bounded by the system physical parameters and it may be relaxed at will to seek a compromise between speed and smoothness of the shifting maneuver. The controllability of the final phase of the synchronization allows for a simplified prediction of the moment when the differential speed will be null or within a defined range. The non-overshooting behavior of the first order system is of uttermost relevance on this phase of the speed synchronization phase. The described strategy may minimize the rotational speed synchronization time, which may allow for a repeatable engagement phase.

[0056] Referring now to FIG. 7, a visualization of an example way to determine when the rotational speed differential is expected or predicted to reach a maximum differential rotational speed where the shifter may be adjusted to engage the on-coming gear. The vertical axis represents a rotational speed difference between a rotational speed of an intermediate shaft and a rotational speed of an on-coming gear (e.g., a gear that is in the process of being engaged). The horizontal axis represents time and time increases from the left side of the plot to the right side of the plot. Trace 702 represents the rotational speed difference between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear. Line 706 represents the expected or predicted rotation speed difference trajectory from a time along trace 702. Horizontal line 704 represents a maximum rotational speed difference between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear where the shifter may adjust the dog clutch to engage the on-coming gear.

[0057] The evolution of the rotational speed difference between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear may not be linear, but it may be linearized in the form of $\omega.\text{sub}.d=b.\text{sub}.0+b.\text{sub}.1t$ and defining $a=\omega.\text{sub}.d$, the dynamic of the differential speed may be written in state-space representation as:

$$[00002] \dot{x} = \begin{pmatrix} \dot{d} \\ \dot{a} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} d \\ a \end{pmatrix} = Ax$$

where A is the state matrix, x is the state vector, and the dot denotes derivative with respect to time.

Defining the output vector y as the actual differential speed as measured by the transmission's sensors, its matrix representation is:

$$[00003] y = \dot{d} = (10) \begin{pmatrix} d \\ a \end{pmatrix} = Cx$$

where C is the output matrix. A state observer may be defined for the system starting from state matrix A and the output matrix C as:

$$[00004] \begin{aligned} \dot{\hat{x}} &= A\hat{x} - L(\hat{y} - y) = A\hat{x} - L(C\hat{x} - y) = (A - LC)\hat{x} + L \dot{d} \\ \hat{y} &= C\hat{x} \end{aligned}$$

where L is the observer gain matrix and the variables with the hat ($\{\text{circumflex over } (\)\}$) are the observer state x and the observer output vector $\hat{y}$. Defining the error as the difference between the system state vector and the observer state vector: $e = x - \hat{x}$, the error dynamic may be described by the equation:

$$[00005] \dot{e} = \dot{x} - \dot{\hat{x}} = A(x - \hat{x}) + L(\hat{y} - y) = Ae + L(C\hat{x} - Cx) = Ae - LCe = (A - LE)e$$

where the error e converges in time to zero provided matrix $(A - LC)$ is definite-negative. The observer gain matrix entries may be chosen such that the error is reasonably small in a time that gives the system state assessment an effectively usable value.

[0058] Once a good system state is available, an estimate of the time t' when the differential speed between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear is null or a defined value $\omega_{sub,d,max}$ (e.g., a maximum rotational speed differential where the on-coming gear may be engaged). The following equation defines a time lapse $(t' - t)$ shown in FIG. 7 to have the differential rotational speed be at the target or requested value $\omega_{sub,d,max}$:

$$[00006] \hat{d}(t) = \dot{d}_{max} = \dot{a}(t) \cdot \text{Math.}(t' - t)$$

[0059] Turning now to FIG. 8, a plan view of example dog clutch teeth is shown. Tooth **802** is from the upper one the sleeve, whereas the teeth **804** and **806** are spline teeth (e.g., **306** of FIG. 3) from either of gear **252** or gear **256** of FIG. 2. The sleeve is spinning at the same speed of gear **258** in FIG. 2, and it ultimately gets its angular velocity from the vehicle wheels while the gears obtain their angular velocity from the electric motor. The difference of the two angular velocities is $\omega_{sub,d}$ as seen and in a tangential direction the teeth of the dog clutch are moving with relative speed $\omega_{sub,d} D_{sub,p} / 2$, where $D_{sub,p}$ is the pitch diameter of the engagement ring spline. The sleeve may have a perpendicular motion (axial movement driven by a shift fork, top to bottom in FIG. 8) to engage the locking ring teeth.

[0060] FIG. 9 is a plan view of the same dog clutch teeth of FIG. 8 and a view of an example trajectory **902** for a tooth **802** of a sleeve to engage spline teeth **804** and **806**. FIG. 9 depicts the start of the engagement stroke S of a transmission gear shift and it defines the origin of the time axis and an example trajectory **902** of the tooth **802**. The engagement may be considered achieved when the undercut flanks of the teeth come in contact or, to put it another way, when the sleeve travels the stroke S , as visible in FIG. 10. The stroke S is not a defined quantity but the space the sleeve travels to land into a reliably engaged position. For a successful engagement, the following two inequalities are satisfied:

$$[00007] \begin{aligned} \int_0^{t^*} v_A(t) dt &= \tilde{v}_A t^* > s \\ \int_0^{t^*} \frac{\dot{d}(t) D_p}{2 dt} &< \frac{p}{2} = \frac{D_p}{Z} \end{aligned}$$

where $v_{sub,A}(t)$ is the speed profile of the shifter actuator, $\{\text{tilde over } (v)\}_{sub,A}$ is the sleeve average speed during the maneuver, p is the pitch of the teeth as measured on the pitch diameter, Z is an actual total number of teeth, and t^* is the time it takes to move the sleeve far enough to achieve a stable torque transfer capacity for the dog clutch. The equation is conservative since the cavity between two teeth is wider than the tooth thickness. The shifter actuator speed and the

differential rotational speed may be defined by the control strategy and they may be constrained by boundary conditions. The actuator speed may be upper constrained by a maximum flow of a valve that controls flow at a valve that controls pressure in the actuator. The maximum rotational differential speed $\omega_{\text{sub.d,max}}$ may be upper constrained to reduce a possibility of shift shock and engagement noise. An upper constraint to the duration of the engagement phase may be derived from the second inequality as:

$$[00008] t^* < \frac{p}{d_{\text{max}} D_p} = \frac{1}{z_{d,\text{max}}}$$

Replacing this inequality with the first inequality provides a direct link between the shifter actuator average speed and the maximum differential speed between the rotational speed of the intermediate shaft and the rotational speed of the on-coming gear that portends successful gear engagement yields:

$$[00009] \tilde{v}_A \frac{p}{s D_p} = \tilde{v}_a \frac{v}{s z} > d_{\text{max}}$$

This inequality may be conservative provided that the speed profile is monotonic. A speed increasing shift actuator assures that the condition is satisfied.

[0061] FIG. **10** is a plan view of the same dog clutch teeth shown in FIG. **8** now shown in a partially engaged position. FIG. **11** is a plan view of the same dog clutch teeth shown in FIG. **8** now shown in a position where the teeth may repel each other as the intermediate shaft and the propulsion source rotate. FIG. **12** is a plan view of the same dog clutch teeth shown in FIG. **8** and an additional tooth **1202** in a position where the teeth may begin to interfere with each other during rotation of the intermediate shaft and the propulsion source. FIG. **13** shows a plan view of two teeth shown in FIG. **8** and forces that may operate on the two teeth. FIG. **14** is a plan view of the same dog clutch teeth shown in FIG. **8** and an additional tooth **1402** in a position where the teeth are fully engaged, thereby fully engaging the on-coming gear.

[0062] FIG. **9** shows a tip **904** of the sleeve tooth well apart from the tip **906** of the gear tooth **804** and in this condition the relative speed may be relevant allowing for the synchronization to take place. A minimum dependable residual distance of two times the tip height h is applied during the synchronization. Tip high $2 \cdot \text{Math.h}(2h)$ is shown in FIG. **12** and in one example it may be 1.5 millimeters. Note that if the sleeve travels this distance, it brings direct contact of the undercut flank of the teeth. From this time onward, the dog clutch has the capability to transfer torque between the sleeve and the on-coming gear even if the sleeve is far from reaching its end-stop. When the sleeve end-stop **311** of FIG. **3** reaches the gear spline tip, the dog clutch achieves the maximum overlap Δ , as shown in FIG. **14**. The Δ is around two millimeters in one example. In the above engagement inequality, the residual stroke s appears at the denominator. Keeping s at a minimum value helps in fulfilling the condition. However, the rate of speed change transient of the gear shift actuator may result in a lower average speed $\{\tilde{v}\}_{\text{sub.A}}$, thereby decreasing a probability of a successful engagement that is enhanced having the peak speed of the gear shifter actuator when the teeth's tips are nearly in contact. For this reason, the engagement operation may begin with the sleeve further apart from the minimal reliable distance.

[0063] The hydraulic gear shifter actuator speed, its rate of speed change transient together with the dependencies on the oil temperature are known and anyhow may be calibrated with a learning function. Sleeve position is known via sensor **216** of FIG. **2**, so as it is known, the distance to be travelled to achieve the engagement may be determined. The engaged position may be calibrated. Therefore, the time lapse that the sleeve needs to travel to reach the tip-to-tip position as shown in FIG. **12** at its maximum speed may be determined. A state observer may be applied to predict in real time the time lapse to achieve the differential speed $\omega_{\text{sub.d,max}}$. When the two time-lapses are identical, the sleeve is launched via operating the shifter actuator in feed forward towards the engaged position.

[0064] The configuration depicted in FIG. **12** is a desired or requested-case scenario and, provided the engagement inequality is satisfied, will result in a successful engagement. The angular position

of the tips is anyhow random and the configuration depicted in FIG. 11 may be much more likely to happen resulting in an interference between the tip flanks. The choice of $\omega_{sub,d,max}$ is such that the relative speed is in the order of tens of millimeters per second to reduce a possibility of degradation, wear, and interference noise. FIG. 13 shows the tips' flanks resting one upon the other immediately after such interference takes place. FIG. 13 also represents the force that the sleeve applies to the engagement ring tooth in the solid line and the reaction to the sleeve in dashed line. The component of the reaction force $F'_{sub,R}$ parallel to the tip flank $F_{sub,S}$ is the friction and it will keep the sleeve in a stalled position as long as $F_{sub,S}$ is smaller than $\mu F_{sub,N}$, with μ being the static friction coefficient. The angle between the resulting force $F_{sub,R}$ and the tip flank depend solely on the ratio between the actuator force $F_{sub,A}$ and the tangential force delivered by the motor $F_{sub,T}$, whereas the ratio of the friction force $F_{sub,S}$ and the normal force $F_{sub,N}$ depends both on the angle and on the tip angle. The engagement is facilitated for sharpest tip and minimum tangential force $F_{sub,T}$; however, a pointed tip is not viable because of increased brittleness of the tooth. Moreover, it is an observed behavior of dog clutches or synchronizers, an increased probability of missed engagement with transmission ageing due to the wear flattening the tooth tip. For this reason, it is desirable to have the tangential force null at the dog ring during the engagement.

[0065] Thus, the system of FIGS. 1-4 and 6-14 provides for an electric propulsion system, comprising: an electric machine; a step gear ratio transmission including at least two gear ratios, the transmission including a shifter actuator; and one or more controllers including executable instructions that cause the one or more controllers to estimate an amount of time for a differential speed between a rotational speed of an on-coming gear and a rotational speed of a transmission shaft to be less than a threshold rotational speed. In a first example, the electric propulsion system includes where threshold rotational speed is a maximum rotational speed difference between the rotational speed of the on-coming gear and the rotational speed of the transmission shaft for engaging the on-coming gear. In a second example that may include the first example, the electric propulsion system includes where the amount of time is determined via an observer. In a third example that may include one or both of the first and second examples, the electric propulsion system further comprises additional executable instructions that cause the one or more controllers to estimate an amount of time to engage the on-coming gear via the shifter actuator based on an average speed of the shifter actuator. In a fourth example that may include one or more of the first through third examples, the electric propulsion system further comprises additional executable instructions that cause the one or more controllers to reduce the differential speed between the rotational speed of the of the on-coming gear and the rotational speed of the transmission shaft via operating the electric machine in a speed control mode and adjusting speed of the electric machine when an off-going gear is disengaged and while the step gear ratio transmission is in a neutral state. In a fifth example that may include one or more of the first through fourth examples, the electric propulsion system includes where speed of the electric machine is adjusted during a speed synchronization phase of a gear shift. In a sixth example that may include one or more of the first through fifth examples, the electric propulsion system includes where the shift actuator is mechanically coupled to a shift fork and a dog clutch, and where the transmission shaft is a transmission output shaft. In a seventh example that may include one or more of the first through sixth examples, the electric propulsion system includes where the transmission shaft is an intermediate shaft.

[0066] Thus, the system of FIGS. 1-4 and 6-14 also provides for an electric propulsion system, comprising: an electric machine; a step gear ratio transmission including at least two gear ratios, the transmission including a shifter actuator; and one or more controllers including executable instructions that cause the one or more controllers to estimate an amount of time for a differential speed between a rotational speed of an on-coming gear and a rotational speed of a transmission shaft to be less than a threshold rotational speed, and additional instructions that cause the one or

more controllers to adjust a position of the shift actuator in response to the amount of time for the differential speed between the rotational speed of the on-coming gear and the rotational speed of the transmission shaft being less than the threshold rotational speed. In a first example, the method further comprises additional executable instructions to operate the electric machine in a speed control mode while adjusting the position of the shift actuator. In a second example that may include the first example, the method includes where the electric machine is operating in a speed control mode during a synchronization phase of a gear shift. In a third example that may include one or both of the first and second examples, the method includes where the transmission shaft is an intermediate shaft, and where the position of the shift actuator is adjusted from a neutral position to a reliably engaged position. In a fourth example that may include one or more of the first through third examples, the method includes where the shift actuator is mechanically coupled to a dog clutch.

[0067] Turning now to FIGS. **15** and **16**, a flowchart of a method for shifting a transmission is shown. The transmission being shifted may be of the type that is shown in FIGS. **1** and **2**. The method of FIGS. **15** and **16** may be incorporated into and may cooperate with the system of FIGS. **1-3**. Further, at least portions of the method of FIGS. **15** and **16** may be incorporated as executable instructions stored in non-transitory memory of one or more controllers while other portions of the method may be performed via the one or more controllers transforming operating states of devices and actuators in the physical world.

[0068] At **1502**, method **1500** operates the vehicle's traction motor or propulsion source in a torque control mode. Torque control mode allows a speed of the traction motor to vary while the torque of the traction motor follows a requested torque value or request. Thus, the traction motor torque is adjusted to follow a requested or commanded torque while the traction motor speed is allowed to vary. Method **1500** proceeds to **1504**.

[0069] At **1504**, method **1500** judges whether or not conditions are present for a transmission gear shift to be performed. In one example, a transmission may be shifted according to a shift schedule that is stored in controller memory. The shift schedule may request shifts based on driver demand torque and present vehicle speed. If method **1500** judges that conditions are present for a gear shift, the answer is yes and method **1500** proceeds to **1506**. Otherwise, the answer is no and method **1500** proceeds to exit.

[0070] At **1506**, method **1500** generates a null torque or zero torque at the dog clutch that controls engagement of the off-going gear. Method **1500** may adjust torque of the traction motor to achieve null or zero torque transfer through the dog clutch of the off-going gear while the traction motor continues to operate in a torque control mode. Method **1500** proceeds to **1508**.

[0071] At **1508**, method **1500** disconnects the off-going gear from the intermediate shaft and/or the transmission output shaft by moving the dog clutch toward its neutral position as shown in FIG. **4** where dog clutch teeth are not engage with spline teeth of the off-going gear or spline teeth of the on-coming gear. Method **1500** proceeds to **1510**.

[0072] At **1510**, method **1500** switches the operating mode of the traction motor from torque control to speed control. In speed control mode, the rotational speed of the traction motor is commanded to follow a speed request or command while the torque of the traction motor is allowed to vary. Method **1500** also adjusts the rotational speed of the traction motor so that a rotational speed of an on-coming gear (e.g., gear that is to be engaged during the shift) matches a rotational speed of an intermediate shaft, or alternatively, to a rotational speed of a transmission output shaft. The transmission gear shifter adjusts a position of the dog clutch sleeve to a position where the dog clutch is reliably disengaged from the off-going gear and the on-coming gear. This allows the speed of the on-coming gear, which is coupled to the traction motor, to be adjusted independent of vehicle speed. In one example, method **1500** adjusts speed of the traction motor so that a differential speed between the on-coming gear and the intermediate shaft or the transmission output shaft is reduced toward a value of zero as shown in FIG. **6**. Method **1500** proceeds to **1512**,

1514, and **1540**.

[0073] At **1512**, method **1500** measured the rotational speed of the propulsion source or motor. The rotational speed may be measured via a position sensor that senses teeth of a gear or via an analog tachometer. Method **1500** proceeds to **1516**.

[0074] At **1514**, method **1500** measured the rotational speed of the transmission shaft transmission, or alternatively, a rotational speed of an intermediate shaft in the transmission. The rotational speed of the intermediate shaft of the transmission may be inferred by dividing the rotational speed of the output shaft of the transmission by a ratio of gears between the transmission intermediate shaft and the transmission output shaft. Alternatively, the intermediate shaft rotational speed may be measured by a sensor. Method **1500** proceeds to **1516**.

[0075] At **1516**, method **1500** determines a differential speed. In one example, method **1500** may determine a differential speed as the variance between the rotational speed of the engagement sleeve (e.g., **802**) and the rotational speed of the teeth (**804/806**). These rotational speeds can be measured either directly or indirectly using speed sensors positioned on a separate interconnected shaft, taking into account the gear ratios. Method **1500** proceeds to **1518** and **1520**.

[0076] At **1518**, method **1500** adjusts or controls the motor speed. In one example, method **1500** may operate the vehicle's traction motor in a speed control mode and adjust motor speed to reduce a differential speed between teeth (e.g., **804/806**) and the corresponding engagement sleeve (e.g., **802**). Method **1500** returns to **1512**.

[0077] At **1520**, method **1500** applies an observer to estimate an amount of time (t') it will take for the differential speed between the on-coming gear and the intermediate shaft, or alternatively the transmission output shaft to be less than or equal to a maximum rotational speed differential between the on-coming gear and the intermediate shaft, or alternatively, the transmission output shaft for engaging the on-coming gear via the dog clutch (e.g., $\omega_{sub.d,max}$). The observer may be as discussed in the description of FIG. 7 and the estimate of the amount of time (t') to reduce the differential speed to the maximum rotational speed differential (e.g., $\omega_{sub.d,max}$) may be determined as discussed with regard to FIG. 7. Method **1500** supplies the amount of time (t') to step **1522**.

[0078] At **1540**, method **1500** determines a position of the transmission gear shift actuator, which is indicative of shift fork position and dog clutch sleeve position since these devices are mechanically coupled together. The position of the transmission gear shift actuator may be determined via a position sensor that supplies a signal that is indicative of transmission gear shift actuator position to a controller. Method **1500** proceeds to **1542**.

[0079] At **1542**, method **1500** estimates or predicts an amount of time it will take to move the shift actuator to a position to engage the on-coming gear. In one example, the amount of time it will take to move the shift actuator from its present position to a position where the on-coming gear is engaged (e.g., t^*) may be determined via solving the equations mentioned in the description of FIG. 7 for t^* . Method **1500** supplies the amount of time to move the shift actuator t^* to step **1522**.

[0080] At **1522**, method **1500** judges whether or not the amount of time (t') it will take for the differential speed between the on-coming gear and the intermediate shaft, or alternatively the transmission output shaft to be less than or equal to a maximum rotational speed differential between the on-coming gear and the intermediate shaft, or alternatively, the transmission output shaft is less than or equal to the amount of time it will take to move the shift actuator from its present position to a position where the on-coming gear is engaged (e.g., t^*). If so, the answer is yes and method **1500** proceeds to **1524**. Otherwise, the answer is no and method **1500** returns to **1512**, **1514**, and **1540**.

[0081] At **1524**, method **1500** commands the shifter actuator to engage the on-coming gear. The on-coming gear may be engaged by moving the dog clutch sleeve to engage spline teeth of the on-coming gear. Method **1500** proceeds to **1526**.

[0082] At **1526**, method **1500** determines or reads the gear shifter actuator position via a position

sensor. Method **1500** proceeds to **1528**.

[0083] At **1528**, method **1500** judges whether or not the gear shifter actuator is at an end of stroke position (e.g., a position where the on-coming gear is reliably engaged). If so, the answer is yes and method **1500** proceeds to **1530**. Otherwise, the answer is no and method **1500** returns to **1524**.

[0084] At **1530**, method **1500** changes the operating mode of the propulsion source or traction motor to a torque control mode and torque of the traction motor is adjusted to the driver demand torque. Method **1500** proceeds to exit.

[0085] Thus, method **1500** adjusts a position of a shift actuator in response to an amount of time estimated for a differential rotational speed between an on-coming gear and a rotational speed of a shaft (e.g., intermediate shaft or transmission output shaft) to be less than a threshold speed (e.g., a maximum rotational speed difference for dog clutch engagement).

[0086] The methods of FIGS. **15** and **16** provide for a method for operating a vehicle, comprising: adjusting a shift actuator in response to a predicted an amount of time for a clutch actuator to move from a first position to a second position being less than or equal to an amount of time for a differential speed to be a predetermined differential speed. In a first example, the method includes where the first position is a position where a dog clutch is disengaged from an off-going gear and where the second position is a reliably engaged position for an on-coming gear. In a second example that may include the first example, the method includes where the amount of time for the differential speed to be the predetermined differential speed is based on output of an observer. In a third example that may include one or both of the first and second examples, the method includes where the predetermined differential speed is a maximum speed differential between a rotational speed of a gear and a rotational speed of a shaft for engaging a dog clutch. In a fourth example that may include one or more of the first through third examples, the method includes where the differential speed is a rotational speed difference between a rotational speed of an on-coming gear and a rotational speed of a shaft. In a fifth example that may include one or more of the first through fourth examples, the method includes where the shaft is an intermediate shaft. In a sixth example that may include one or more of the first through fifth examples, the method further comprises operating an electric machine in a speed control mode to reduce the differential speed while adjusting the shift actuator.

[0087] Note that the example control and estimation routines included herein may be used with various powertrain and/or vehicle system configurations. The control methods and routines disclosed herein may be stored as executable instructions in non-transitory memory and may be carried out by the control system including the controller in combination with the various sensors, actuators, and other transmission and/or vehicle hardware. Further, portions of the methods may be physical actions taken in the real world to change a state of a device. Thus, the described actions, operations and/or functions may graphically represent code to be programmed into non-transitory memory of the computer readable storage medium in the vehicle and/or transmission control system. The specific routines described herein may represent one or more of any number of processing strategies such as event-driven, interrupt-driven, multi-tasking, multi-threading, and the like. As such, various actions, operations, and/or functions illustrated may be performed in the sequence illustrated, in parallel, or in some cases omitted. Likewise, the order of processing is not necessarily required to achieve the features and advantages of the examples described herein, but is provided for ease of illustration and description. One or more of the illustrated actions, operations and/or functions may be repeatedly performed depending on the particular strategy being used. One or more of the method steps described herein may be omitted if desired.

[0088] While various embodiments have been described above, it is to be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant arts that the disclosed subject matter may be embodied in other specific forms without departing from the spirit of the subject matter. The embodiments described above are therefore to be considered in all respects as illustrative, not restrictive. As such, the configurations and routines

disclosed herein are exemplary in nature, and that these specific examples are not to be considered in a limiting sense, because numerous variations are possible. For example, the above technology may be applied to electric vehicles and hybrid vehicles including induction and synchronous electric machines. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed herein.

[0089] The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. These claims may refer to “an” element or “a first” element or the equivalent thereof. Such claims may be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Other combinations and sub-combinations of the disclosed features, functions, elements, and/or properties may be claimed through amendment of the present claims or through presentation of new claims in this or a related application. Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

Claims

1. An electric propulsion system, comprising: an electric machine; a step gear ratio transmission including at least two gear ratios, the step gear ratio transmission including a shifter actuator; and one or more controllers including executable instructions that cause the one or more controllers to estimate an amount of time for a differential speed between a rotational speed of an on-coming gear and a rotational speed of a transmission shaft to be less than a threshold rotational speed.
2. The electric propulsion system of claim 1, where threshold rotational speed is a maximum rotational speed difference between the rotational speed of the on-coming gear and the rotational speed of the transmission shaft for engaging the on-coming gear.
3. The electric propulsion system of claim 2, where the amount of time is determined via an observer.
4. The electric propulsion system of claim 3, further comprising additional executable instructions that cause the one or more controllers to estimate an amount of time to engage the on-coming gear via the shifter actuator based on an average speed of the shifter actuator.
5. The electric propulsion system of claim 4, further comprising additional executable instructions that cause the one or more controllers to reduce the differential speed between the rotational speed of the of the on-coming gear and the rotational speed of the transmission shaft via operating the electric machine in a speed control mode and adjusting speed of the electric machine when an off-going gear is disengaged and while the step gear ratio transmission is in a neutral state.
6. The electric propulsion system of claim 5, where speed of the electric machine is adjusted during a speed synchronization phase of a gear shift.
7. The electric propulsion system of claim 1, where the shifter actuator is mechanically coupled to a shift fork and a dog clutch, and where the transmission shaft is a transmission output shaft.
8. The electric propulsion system of claim 1, where the transmission shaft is an intermediate shaft.
9. A method for operating a vehicle, comprising: adjusting a shift actuator in response to a predicted an amount of time for a clutch actuator to move from a first position to a second position being less than or equal to an amount of time for a differential speed to be a predetermined differential speed.
10. The method of claim 9, where the first position is a position where a dog clutch is disengaged from an off-going gear and where the second position is a reliably engaged position for an on-coming gear.
11. The method of claim 10, where the amount of time for the differential speed to be the predetermined differential speed is based on output of an observer.
12. The method of claim 11, where the predetermined differential speed is a maximum speed

differential between a rotational speed of a gear and a rotational speed of a shaft for engaging the dog clutch.

13. The method of claim 9, where the differential speed is a rotational speed difference between a rotational speed of an on-coming gear and a rotational speed of a shaft.

14. The method of claim 13, where the shaft is an intermediate shaft.

15. The method of claim 9, further comprising operating an electric machine in a speed control mode to reduce the differential speed while adjusting the shift actuator.

16. An electric propulsion system, comprising: an electric machine; a step gear ratio transmission including at least two gear ratios, the step gear ratio transmission including a shifter actuator; and one or more controllers including executable instructions that cause the one or more controllers to estimate an amount of time for a differential speed between a rotational speed of an on-coming gear and a rotational speed of a transmission shaft to be less than a threshold rotational speed, and additional instructions that cause the one or more controllers to adjust a position of the shifter actuator in response to the amount of time for the differential speed between the rotational speed of the on-coming gear and the rotational speed of the transmission shaft being less than the threshold rotational speed.

17. The electric propulsion system of claim 16, further comprising additional executable instructions to operate the electric machine in a speed control mode while adjusting the position of the shifter actuator.

18. The electric propulsion system of claim 17, where the electric machine is operating in the speed control mode during a synchronization phase of a gear shift.

19. The electric propulsion system of claim 16, where the transmission shaft is an intermediate shaft, and where the position of the shifter actuator is adjusted from a neutral position to a reliably engaged position.

20. The electric propulsion system of claim 16, where the shifter actuator is mechanically coupled to a dog clutch.
